MIL-E-1/1043D
3 October 1977
SUPERSEDING
MIL-E-1/1043C
30 June 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 5876A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pencil triode; glass and metal construction

See figure 1

Mounting position: Any

Weight: 0.85 ounce nominal

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ib	Ic	Pp	Pi
Unit:	V	Vdc	Vdc	mAdc	mAdc	W (See note 1)	W
Class A amp:	6.3 ± 10%	300	-100	25	--	6.25	----
Class C teleg:	6.3 ± 10%	360	-100	25	8	6.25	9.0
Class C teleph:	6.3 ± 10%	275	-100	22	8	4.25	6.0
Freq mult:	6.3 ± 10%	330	-100	22	8	6.25	7.25
TEST CONDITIONS:	6.3	250	Note 18	--	--	----	----

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ehk	Rk	tk	Mod	T (Seal) (See note 1)	Barometric pressure reduced
Unit:	Vdc (See note 2)	Ohms	sec (min)	---	°C	mmHg
Class A amp:	+90	--	60	---	175	55
Class C teleg:	±90	--	60	---	175	55
Class C teleph:	±90	--	60	Anode	175	55
Freq mult:	±90	--	60	---	175	55
TEST CONDITIONS:	---	75 (See note 3)	180	---	---	--

GENERAL:

Qualification - Required

Ⓓ Denotes changes

5876A

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	See notes 4 and 16						
1256	Electrode current (1) (anode)		0.65	II	Ib(1)	12.5	23.5	mAdc
1306	Transconductance		0.65	II	Sm	5400	8100	µmhos
1256	Electrode current (2) (anode)	Rk = 0 Ec = -12 Vdc	0.65	II	Ib(2)	----	55	µAdc
1266	Total grid current	Rk = 0 Ec = -2.5 Vdc	0.65	II	Ic	----	-1.0	µAdc
1231	Emission	Rk = 0 Eb = Ec/Ik = 30 mAdc Ef = 5.5 V	0.65	II	Es	----	10	Vdc
1236	Power oscillation	Eb = 200 Vdc Rk = 0 Rg/Ib = 18 mAdc (max) F = 1,700 ±15 MHz Use cavity per drawing 154-JAN of approved equivalent	0.65	II	Po	285	---	mW
1331	Direct-interelectrode capacitance	Use cavity per drawing 155-JAN or approved equivalent; See note 19	0.65	II	Cgp Cgk	1.2 2.0	1.6 2.6	pF pF
	Quality conformance inspection, part 2							
1301	Heater current		----	---	If	127	143	mA
1031	Low-frequency vibration	Rk = 0 Ec = -2.5 V Rp = 10,000 Ebb = 250 Vdc	----	---	Ep	---	100	mVac
1336	Heater-cathode leakage		----	---	±Ihk	---	50	µAdc
1331	Direct-interelectrode capacitance (3)	Use cavity per drawing 155-JAN or approved equivalent; See note 19	----	---	cpk	---	.035	pF
1031	High-frequency vibration	No voltages See note 5	---	---	---	---	----	----
1316	Amplification factor		---	---	Mu	41	71	----
1211	Insulation of electrodes	Ef = 6.3 E(g-all) = -100 Vdc, (grid negative)	---	---	Rg-all	25	---	Meg
		Ef = 6.3 E(p-all) = -300 Vdc, (anode negative)	---	---	Rp-all	25	---	Meg
1256	Electrode current (3) (anode)	Rk = 0 Ec = -5 Vdc	---	---	Ib(3)	0.5	---	mAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
(D) 1506	Heater-cycling life test	Group D Ef = 6.3 V Ec = Eb = Ehk = 0 See notes 10 and 20	----	----	---	---	---	----
----	Heater-cycling life test end points:							
(D) ----	Inoperatives	See note 11	----	----	---	---	---	----
(D) 1211	Insulation of electrodes		----	----	Rg-all	25	---	Meg
(D) 1336	Heater-cathode leakage		----	----	±Ihk	---	100	µAdc
(D) 1516	Stability life test (1 hour)	Eb = Vary for Pp = 2.0 w (min) 2.5 w (max) TA = room; Ec = 0; Rk = 20 ohms See notes 14 and 17	----	----	---	---	---	----
----	Stability life-test end point:							
1306	Change in trans-conductance in individual tubes		----	----	ΔSm	---	-15	%
(D) 1521	Survival rate life (50 hours)	Stability life-test (1 hour) conditions or equivalent; See notes 12, 13 and 14	----	II	---	---	---	----
----	Survival rate life-test end points:	See note 15	----	----	---	---	---	----
(D) ----	Inoperatives	See note 11	1.0	----	Inop	---	---	----
(D) 1256	Electrode current (2) (anode)		1.0	----	Ib(2)	---	100	µAdc
1236	Power oscillation		1.0	----	Po	200	---	mW
(D) 1501	Intermittent life test	Eb = 300 Vdc Rg/Ib = 25 mAdc Ic = 12 mAdc (max) Ehk = +100 Vdc F = 525 ±25 MHz t = 500 hours Group A	----	----	---	---	---	----
----	Intermittent life test end points:	See note 15						
----	Inoperatives	See note 11	----	----	Inop	---	---	----
1256	Electrode current (2) (anode)		----	----	Ib(2)	---	150	µAdc
1236	Power oscillation		----	----	Po	200	---	mW
(D) 1002	Barometric pressure reduced	Pressure = 55 ±5mmHg Voltage = 500 Vac See notes 6 and 7	----	----	---	---	---	----
(D) ----	Seal fracture	No voltages See notes 6 and 8	----	----	---	30	---	lbs
(D) ----	Torque	No voltages See notes 6 and 9	----	----	---	12.5	---	in-lbs
(D) 1116	Lead fatigue	No voltages See note 6	----	----	---	4	---	arcs

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NOTES:

1. The tube will be provided with adequate cooling in order to maintain the seal temperature at less than 175°C.

2. The reference point for heater-cathode potential shall be the positive terminal of the cathode resistor.

3. The reactance of the cathode by-pass capacitor shall be less than 3 ohms.

4. All tests listed under Quality Conformance Inspection, part 1, are to be performed at the conclusion of the holding period.

5. The tube shall be vibrated in a plane perpendicular to the major axis. No voltages. Freq. = 40 to 60 Hz; t = 3 minutes; G = 10. At the conclusion of the vibration period, the tube shall meet the specified limits for +Ihk, Ib(2), Sm, and low-frequency vibration.

Ⓓ 6. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made a part of Quality Conformance Inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

7. The specified voltage shall be applied between the anode and the grid plus cathode.

Ⓓ 8. Tube supported at cathode and anode sleeves at points 15/16 ±1/64 inch apart with the grid disc centered. Force applied to grid disc without shock perpendicular to axis of tube until fracture occurs. Electrical rejects may be used.

9. The cathode terminal shall be rigidly clamped and a torque of 12.5 in. lbs shall be applied gradually about the tube axis to the anode terminal. Tube shall not fracture. This is a destructive test. Electrical rejects may be used.

Ⓓ 10. A complete cycle shall consist of a 6-minute period as follows: 3 minutes "on", 3 minutes "off." Mechanical or electrical rejects may be used.

11. An inoperative as referenced in post-heater-cycling life test end points, post-survival-rate life test end points, and post-intermittent life test end points shall be defined as a tube having one or more of the following defects: discontinuity or permanent short except tubes shall not be tapped, air leak (Ic in excess of -2.0 µAdc), or Po less than 80 mW.

12. Intermittent cycling of voltages shall be used in accordance with Test Method 1501.

13. Survival-rate life test:

    (a) Operate survival-rate life test sample for 50 hours under conditions specified.

    (b) Tubes shall be tested at the end of this period for the characteristics specified, MIL-STD-105, Inspection Level II, to apply.

    (c) Determine the number of defective tubes at the end of the 50 hour period.

    (d) If more than the allowable number of defectives occur, declare the lot nonconforming.

    (e) An inoperative for this test shall be defined as a tube having one or more of the defects listed in note 11.

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TES: (Cont'd)

14. Survival-rate and stability-life test sample: Life-test samples shall be selected from a lot at random in such a manner as to be representative of the lot. If such a selection results in a sample containing tubes which are outside the initial specification sheets limits, such tubes shall be replaced by randomly selected acceptable tubes. This shall be a nondestructive test.

15. Order of evaluation of life-test defects: If a tube is defective for more than one attribute characteristics, the characteristic appearing first in the life test end point shall constitute the failure.

D 16. "Characteristics within this group shall be sampled in accordance with MIL-STD-105, inspection level II, and shall, excluding inoperatives, Po, Cap(1), Cap(2), meet a combined AQL of 1.0 percent. The AQL for individual characteristics is 0.65 percent. A lot failing the 1 percent combined AQL because of defects for one characteristic may be detailed for that characteristic alone. However, if the lot failure results from defects of more than one characteristic, detailing shall include all characteristics in the group. Use single, double, or multiple sampling provisions per MIL-STD-105."

D 17. "Stability life test":

(a) Serially mark all tubes from the sample.
(b) Record referenced characteristic measurements.
(c) Operate for 1 hour (plus 30 minutes minus 0 minutes).
(d) Record referenced characteristic measurements at the end of this life test period.
(e) A defective shall be defined as a tube having a change in referenced characteristics greater than specified on the tube specification sheet.

Change in percentage referenced characteristic =

$$\frac{\text{Initial value minus one hour value}}{\text{Initial value}} \times 100$$

18. Whenever E_c is not specified on any test, the grid shall be grounded directly or through a grid resistor (if specified) except where related MIL-STD-1311 test methods imply grid variance to accomplish the test.

D 19. A capacitance bridge operating frequency of approximately 1 to 2 kHz may be substituted in lieu of approximately 500 kHz.

D 20. Test may be performed on a Group D basis every 90 days with the lot consisting of the current month and two previous months. Failure of any tube shall revert back to Group D monthly lot sampling. After three consecutive successful lot submissions have been completed, the test may revert to the quarterly basis.

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Navy - EC
Air Force - 85

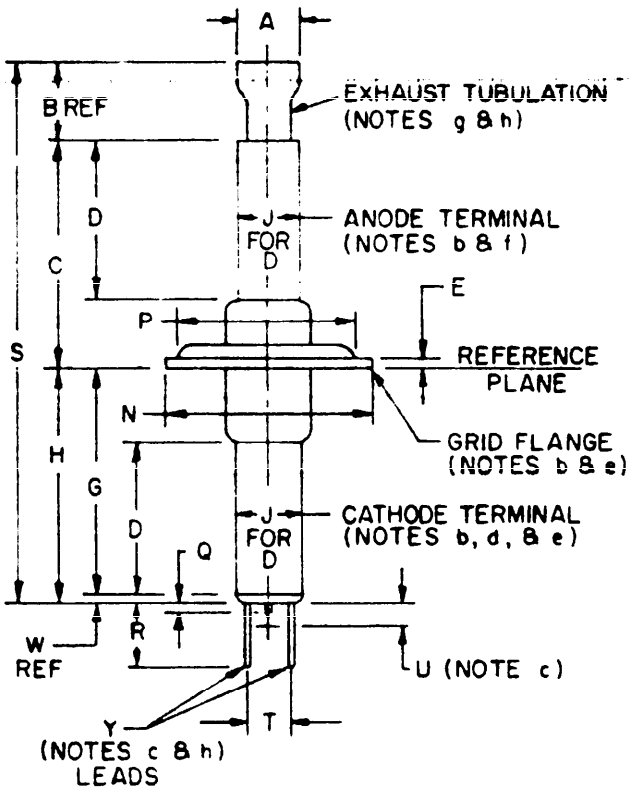
Review activities:
Air Force - 99
DLA - ES

User activities:
Army - MU
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Air Force - 85

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONF. INSP. PT.3, (PERIODIC CK. NOTE a)				
A	-	.230	-	5.84
B	-	.488	-	12.40
C	.835	.885	21.21	22.47
D	.580	-	14.73	-
E	.029	.035	.74	.88
G	.695	-	17.65	-
H	.720	.865	18.29	21.97
J	.245	.255	6.22	6.47
N	.807	.817	20.50	20.75
P	-	.687	-	17.44
Q	-	.040	-	1.01
R	.200	.240	5.08	6.09
S	-	2.043	-	51.89
T	.075	.155	1.91	3.93
U	-	.080	-	2.03
W	-	.025	-	.63
Y	.018	.022	.46	.55

ELECTRICAL CONNECTIONS

CONTACT SURFACE	ELEMENT
DIM J	a & k
DIM N	g
LEADS	h-h

NOTES:

a. THESE DIMENSIONS SHALL BE INSPECTED PERIODICALLY EVERY THREE MONTHS DURING PRODUCTION. AT START OF PRODUCTION, A SAMPLE OF SIX TUBES SELECTED AT RANDOM SHALL MEET ACCEPTANCE REQUIREMENTS WITH NO ALLOWABLE DEFECTS. FAILURE OF ANY DIMENSION TO MEET ACCEPTANCE REQUIREMENTS SHALL CAUSE THAT DIMENSION TO BE INSPECTED WITH AN AQL OF 2.5 PERCENT INDIVIDUAL AND 4.0 PERCENT COMBINED. WHEN THREE SUCCESSIVE SAMPLES AS SPECIFIED ABOVE MEET ACCEPTANCE REQUIREMENTS FOR THIS DIMENSION, THE DIMENSION WILL REVERT TO PERIODIC INSPECTION.

b. ECCENTRICITY OF GRID FLANGE OF ANODE SHALL BE .010 MAX WITH RESPECT TO CATHODE TERMINAL. ECCENTRICITY OF ANODE SHALL BE GAUGED BY CHUCKING CATHODE TERMINAL .050 MIN AND .100 MAX FROM CATHODE-GLASS SEAL AND GAUGING 1/2 TIR .080 FROM END OF ANODE TERMINAL. ECCENTRICITY OF GRID FLANGE SHALL BE GAUGED BY CHUCKING CATHODE TERMINAL AS SPECIFIED ABOVE AND GAUGING 1/2 TIR OF EDGE OF GRID FLANGE. AQL SHALL BE 6.5 PERCENT, INSPECTION LEVEL S3

c. LEADS SHALL BE TRIMMED AND SHALL BE TINNED EXCEPT AS SPECIFIED BY DIMENSION U. UNTINNED LENGTH OF LEADS SHALL BE MEASURED FROM END OF CATHODE TERMINAL. PERIODIC CHECK SHALL APPLY.

d. GLASS FROM STEM SEALING SHALL NOT BE PRESENT ON CATHODE TERMINAL SURFACE. GLASS BUTTON SHALL NOT EXTEND RADIALLY BEYOND CATHODE TERMINAL SURFACE. PERIODIC CHECK SHALL APPLY.

e. ALLOWABLE TILT OF GRID FLANGE SHALL BE .020 MAX WITH RESPECT TO AXIS OF CATHODE TERMINAL AND SHALL BE DETERMINED BY CHUCKING CATHODE TERMINAL, ROTATING TUBE THROUGH ONE COMPLETE REVOLUTION; AND GAUGING TOTAL TRAVEL DISTANCE OF FLANGE PARALLEL WITH AXIS OF CATHODE TERMINAL AND .020 FROM EDGE OF FLANGE. AQL SHALL BE 6.5 PERCENT, INSPECTION LEVEL S3.

f. DIAMETER OF ANODE TERMINAL SHALL NOT BE INCREASED BY SOLDER OR MELTED PLATING FROM EXHAUST TUBULATION. PERIODIC CHECK SHALL APPLY.

g. ALLOWABLE BURNED AREA AT EXHAUST TUBULATION END OF ANODE TERMINAL SHALL BE .0025 SQUARE INCH MAXIMUM. AQL SHALL BE 6.5 PERCENT, INSPECTION LEVEL S3

h. ALL EXTERNAL METAL SURFACES, EXCEPT LEADS AND EXHAUST TUBULATION, SHALL BE SILVER OR GOLD PLATED 50 mg/in^2 MINIMUM. PLATING SHALL NOT BE BLISTERED AND PLATED AREAS SHALL EXHIBIT NO EXPOSED BASE METAL. PERIODIC CHECK OF NOTE 6 SHALL APPLY.

FIGURE 1 Outline drawing of electron tube type 5876A

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-E-1/1043D Electron Tube, Negative Grid (Microwave) Type 5876A

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

FSC 5960

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